

# Manar Salah

Flutter Developer

✉ manarsalah3200@gmail.com    ☎ +201109057028    📍 Cairo, Egypt    🔗 LinkedIn  
🐙 GitHub    📁 Portfolio

## SUMMARY

**Mobile Software Engineer** experienced in the full mobile lifecycle—building, testing, and shipping scalable Flutter apps with Clean Architecture. Collaborative, remote-ready, and dedicated to writing clean, maintainable code within a high-performing engineering team.

## PROJECTS

### ZooVerse — Multi-Role Zoo Management & Visitor Platform (🔗 Case Study) Jun. 2026

**Technologies:** Flutter, Dart, Firebase, Cloud Firestore, Supabase, BLoC/Cubit, Clean Architecture, GoRouter, Dependency Injection, Geofencing, OpenStreetMap, Google Maps, Paymob Payment Gateway, Real-time Systems, Localization, Responsive Web

- Delivered **3 integrated applications** — a Visitor mobile app, an Employee mobile app with role-based workflows, and an Admin web dashboard — as a single zoo management ecosystem on a shared backend.
- Delivered core visitor journeys covering animal discovery, **Google Maps navigation**, and QR-based ticket entry with **Paymob payment integration**.
- Engineered real-time animal assignment using **Firestore transactions**, preventing conflicting claims between multiple keepers.
- Established a full emergency-response lifecycle with priority levels, evidence capture, and status tracking from Pending to Resolved.
- Engineered employee workflows for ticket validation, **duplicate-entry prevention**, and **geofenced attendance tracking**.
- Implemented animal-care reporting with feeding records, evidence uploads, and critical-condition escalation to administrators.

### Graduation Assistant — AI-Powered Graduation Project Workspace Mar. 2026

**Technologies:** Flutter, Dart, BLoC, Clean Architecture, Gemini AI, Supabase, Hive

- Developed an AI-powered workspace that turns raw project ideas into structured deliverables — SRS, ERD, use cases, and risk analysis.
- Architected a **local-first data layer with Hive**, backed by bidirectional Supabase synchronization and conflict resolution.
- Integrated **Google Gemini** with prompt sanitization, structured JSON parsing, and rate-limited retry handling.
- Built a custom 2D diagram renderer using **Flutter Canvas** to generate interactive ERD and Use Case diagrams with Bézier curve routing — no third-party library.
- Secured data ownership end-to-end with Supabase Auth, PostgreSQL, and **Row-Level Security (RLS)**.

### Tabibi — Multi-Role Healthcare Management Platform (🔗 Case Study) Nov. 2025

**Technologies:** Flutter, Dart, Firebase, Cloud Firestore, BLoC/Cubit, Clean Architecture, GoRouter, Dependency Injection, Firebase Notifications, PDF Generation, Localization, Responsive Web, Custom Design System

- Delivered **3 integrated applications** — a Patient app, a Doctor app, and an Admin web dashboard — as a single healthcare platform on a shared real-time backend.
- Implemented the full appointment lifecycle, from booking and availability to cancellation and follow-up scheduling.
- Enabled **real-time doctor-patient messaging** with text, files, voice notes, and read receipts.
- Produced digital clinical records with diagnosis, prescriptions, and **PDF medical reports**.
- Implemented doctor access control with admin-generated invitation codes and role-based permissions.
- Established shared architecture across all 3 apps using **Clean Architecture** and reusable design-system components.

**Technologies:** Flutter, Dart, MediaPipe, TensorFlow Lite, On-Device ML

- Built an **offline-first Arabic Sign Language app** with bidirectional translation: Sign → Text and Text → Sign, running entirely on-device with no cloud dependency.
- Achieved recognition support for **31 Arabic sign classes** through a real-time computer vision pipeline using MediaPipe hand landmarks and a custom TFLite classifier.
- Implemented temporal prediction stabilization to smooth frame-by-frame noise and increase recognition consistency in live video input.

**EDUCATION**

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**Obour Institutes — Bachelor of Computer Science**

**October 2022 – June 2026**

Cairo, Egypt | Grade: Good

**TECHNICAL SKILLS**

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**Mobile Development:** Flutter, Dart, Android, Responsive UI

**Architecture & State Management:** Clean Architecture, BLoC/Cubit, Dependency Injection, Repository Pattern

**Backend & APIs:** Firebase, Supabase, PostgreSQL, REST APIs, Row-Level Security (RLS)

**AI & Machine Learning:** Google Gemini API, TensorFlow Lite, MediaPipe, Computer Vision

**Data & Synchronization:** Hive, Offline-First Architecture, Cloud Synchronization, Conflict Resolution

**Tools & Testing:** Unit Testing, Git, GitHub, Debugging

**UI & Internationalization:** Material 3, Arabic/English Localization, RTL Support

**Soft Skills:** Problem-Solving, Attention to Detail, Adaptability, Time Management, Self-Directed Learning, Ownership

**LANGUAGES**

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**Arabic:** Native | **English:** Fluent